Sequence Position Explains an Apparent Complexity Gradient in Mixture-of-Experts Routing Entropy

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Abstract

Routing entropy, the Shannon entropy of the distribution a mixture-of-experts (MoE) router places over its experts at a token, is often averaged over a prompt and read as a measure of how hard the model is working. We show that this average inherits a positional effect that can masquerade as a complexity effect. On a 168-prompt suite graded into twelve levels of intended cognitive complexity, all-token mean routing entropy rises with level in two MoE models (DeepSeek V3.1, Spearman $\rho = +0.80$; Qwen3.5-397B, $\rho = +0.62$; $n = 168$ each) and tracks prompt token count more tightly still ($\rho = +0.88$ and $+0.78$). Per-token routing entropy rises with position inside the prefill in every prompt of a per-token diagnostic, so longer prompts, which the grading also makes longer, pool more high-entropy positions and lift the mean. A readout at the final prefill token, a single position whose causal context is the whole prompt, removes the gradient ($\rho = +0.02$, $p = 0.82$, and $\rho = -0.06$, $p = 0.42$) and reverses the extremes: the short level-1 prompts show slightly higher final-token entropy than the long level-12 prompts in both models ($p < 10^{-3}$). The all-token pattern replicates on DeepSeek R1. In this setting the aggregate gradient is a function of token count and token position, so a between-prompt difference in averaged routing entropy is interpretable as a difference in what the prompts demand only when position or length is controlled; a position-controlled readout is the minimal control.

1 Introduction

A mixture-of-experts (MoE) layer routes each token through a learned gate to a few of many expert subnetworks, and the gate's output is an inspectable distribution over experts at every token. Its Shannon entropy, the routing entropy, is cheap, intrinsic, and locally interpretable: a peaked distribution means the router has committed to a few experts, a flat one means it is spreading weight. Reading a flat distribution as the model "working harder" is a natural extension of that local meaning, and several lines of work treat router confidence as a difficulty signal (Huang et al., 2024; Raposo et al., 2024). Between-prompt comparisons of averaged routing entropy therefore invite a one-line summary such as "harder prompts route more diffusely."

The averaging step is where that reading becomes fragile. A per-token statistic becomes a per-prompt statistic only after a choice of how to pool positions, and the mean over all prefill positions depends on prompt length by construction. Under causal attention each position is computed over every earlier position, so representations, and the router's response to them, can drift systematically with position. If per-token routing entropy rises with position, then the all-token mean of a long prompt pools more high-entropy positions than that of a short one, and prompt length can stand in for whatever the prompts were meant to vary.

We show that this is what happens on a graded complexity suite. Averaged over prefill tokens, routing entropy climbs across twelve levels of intended complexity in DeepSeek V3.1 and Qwen3.5-397B, and it climbs with prompt token count more tightly. Per-token entropy rises with prefill position within prompts. Read at the final prefill token, a single position that has attended to the entire prompt, the gradient with level disappears in both models and the extreme levels swap order. The result concerns one suite, two model families with a replication, and one aggregation choice, and it does not show that routing entropy carries no information about content. Its implication is nonetheless general: a between-prompt gradient in averaged routing entropy is weak evidence on its own that experts respond to intended complexity, because the same gradient is produced by token position, and any workload or difficulty reading of averaged routing entropy needs a position-controlled readout or an explicit model of length and position before it can be interpreted.

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2 Background

Routing and its summary statistics. Sparse MoE layers route each token to a small subset of experts (Jacobs et al., 1991; Shazeer et al., 2017), a design scaled into large transformers by Switch Transformers (Fedus et al., 2022) and, in the models studied here, by the fine-grained routing of DeepSeek (Dai et al., 2024; DeepSeek-AI, 2024, 2025) and the Qwen3 lineage (Yang et al., 2025). Because the router emits a distribution at every token, summary statistics of that distribution are a common readout, going back at least to the per-token analysis of Mixtral (Jiang et al., 2024).

Routing confidence as a difficulty signal. Huang et al. (2024) activate more experts for inputs the router is less confident about, on the premise that harder inputs warrant more computation, and mixture-of-depths (Raposo et al., 2024) routes a per-token compute budget on a related premise. The router responds to input; we show that a between-prompt gradient in averaged routing entropy can be produced by length and position alone.

Position and length as confounds. Causal self-attention (Vaswani et al., 2017) makes a token's representation depend on every earlier position, and transformers behave differently across positions, for instance the attention sinks at initial positions documented by Xiao et al. (2024). That length can stand in for an intended factor is a familiar hazard elsewhere: in reinforcement learning from human feedback, a length-only proxy reproduces much of an apparent quality gradient (Singhal et al., 2024). A companion analysis by the present author separates the prefill and generation passes for a different routing question, expert-domain specialization (Shorthill, 2026); the capture and framing are shared and the measured quantity differs.

3 Methods

Models and capture. DeepSeek V3.1 (DeepSeek-AI, 2024) has 256 experts with 8 active per token across 58 MoE layers; Qwen3.5-397B-A17B (Yang et al., 2025) has 512 experts with 10 active across 60 MoE layers. Both were run from 2-bit dynamic quantized weights (Unsloth `UD-Q2_K_XL`) on a llama.cpp fork (build `b8123`, `capture_activations`) with greedy decoding, and routing was captured in prefill-only mode (`n_predict` $= 0$: the prompt is processed and no token is generated). A DeepSeek R1 run (DeepSeek-AI, 2025) on the same suite serves as a replication. The router output at each prefill token and each MoE layer is the input to every measurement below.

Prompt suite. The suite holds 168 prompts, 14 at each of 12 levels. The levels are an intended-complexity ordering, from rote repetition and factual recall through logical reasoning, analogy, theory of mind, and ethical dilemmas, to self-referential and persona-style prompts. They also vary in length: the low levels are short (level-1 mean 31.7 tokens) and several high levels run long (the longest level, 8, averages 152.6 tokens; level 12 averages 136.9). That length structure is the confound this paper isolates.

Routing entropy and readouts. At a token, each MoE layer's router yields weights over its experts; we take the Shannon entropy of that weight distribution, average across the model's MoE layers, and normalize to $[0, 1]$ by the maximum attainable entropy, giving a per-token routing entropy. We summarize a prompt two ways. The *all-token mean* averages the per-token value over every prefill position. The *final-token readout* takes the value at the last prefill position only; under causal attention that position has attended to the entire prompt, so it is a single-position summary, independent of prompt length by construction.

Position diagnostic. A per-token *position diagnostic* on Qwen3.5-397B recorded the full per-position entropy trajectory for five prompts spanning levels 1 to 9 and 24 to 180 tokens.

Statistics. For each model we compute Spearman rank correlations over all $n = 168$ prompts between routing entropy and (i) the intended level and (ii) the prompt token count, for both readouts. We compare the lowest and highest levels (L1 and L12, $n = 14$ each) with a two-sided Wilcoxon rank-sum test, reporting the standardized statistic z. For the position diagnostic we report each prompt's ordinary-least-squares slope of entropy on position and the Spearman correlation between position and entropy. The level correlation is the quantity a complexity reading would lean on; the token-count correlation is the confound; the final-token readout is the position-controlled check.

4 Results

4.1 The all-token mean rises with intended level, and more tightly with token count

Averaged over all prefill tokens, routing entropy increases across the twelve levels in both models (Figure 1, circles): Spearman $\rho = +0.80$ ($p \approx 5.6 \times 10^{-39}$) for DeepSeek V3.1 and $\rho = +0.62$ ($p \approx 5.7 \times 10^{-19}$)

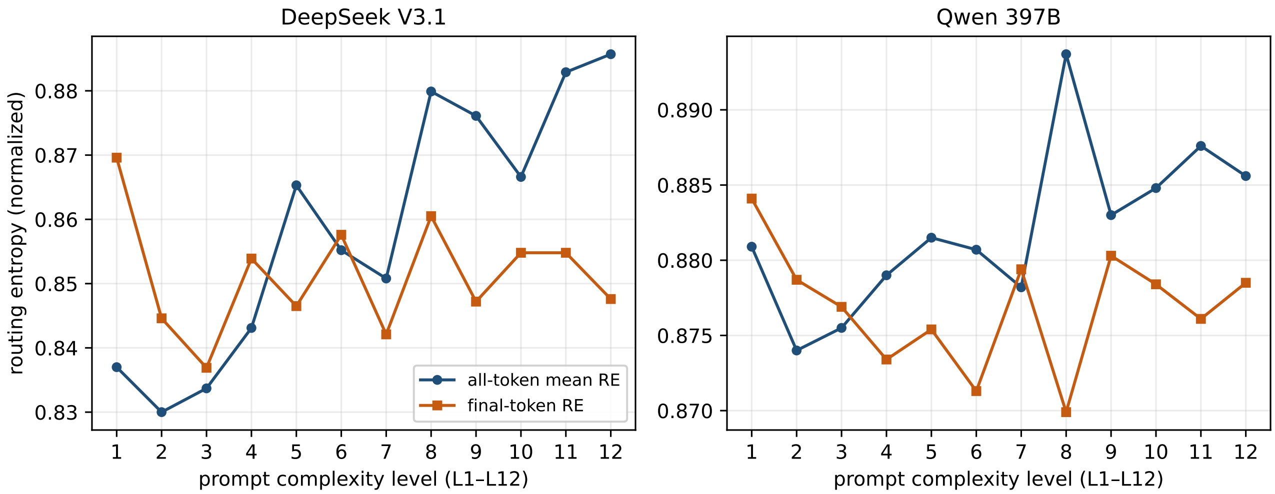

Figure 1: Per-level mean routing entropy against intended complexity level for DeepSeek V3.1 (left) and Qwen 397B (right). Circles: all-token mean. Squares: final-token readout. Each point averages 14 prompts; 168 prompts per model. The all-token mean climbs with level in both models; the final-token readout stays within a narrow band with no monotonic trend.

Table 1: Spearman correlations of routing entropy with intended level and with prompt token count, by readout and model ($n = 168$). The all-token mean orders by level and by length; the final-token readout does neither.

Model	Readout	vs. level (ρ, p)	vs. token count (ρ, p)
DeepSeek V3.1	all-token	$+0.80$ (5.6×10^{-39})	$+0.88$ (1.8×10^{-55})
DeepSeek V3.1	final-token	$+0.02$ (0.82)	$+0.16$ (0.037)
Qwen 397B	all-token	$+0.62$ (5.7×10^{-19})	$+0.78$ (8.3×10^{-36})
Qwen 397B	final-token	-0.06 (0.42)	-0.22 (0.0042)

for Qwen (Table 1). Read alone, this is the gradient a complexity proxy predicts. The same average follows prompt length more closely: across the 168 prompts it correlates with token count at $\rho = +0.88$ ($p \approx 1.8 \times 10^{-55}$) and $\rho = +0.78$ ($p \approx 8.3 \times 10^{-36}$). Figure 2 shows the per-level means against per-level mean token count for DeepSeek V3.1: the entropy ordering and the length ordering largely coincide. Because token count and level are themselves strongly related in this suite, a measure that rises with length also rises with level.

4.2 Per-token routing entropy rises with prefill position

The position diagnostic makes the mechanism visible (Figure 3). In every one of the five Qwen3.5-397B prompts, per-token routing entropy rose with position: ordinary-least-squares slopes were positive with $p \leq 0.012$ in each prompt (2.6×10^{-3} per token for the 24-token prompt down to 1.1×10^{-4} for the 180-token prompt), and Spearman correlations between position and entropy ranged from $+0.22$ to $+0.56$ (all $p < 0.02$). The rise is front-loaded: the first five positions, which share the chat template prefix, average 0.833 in every prompt, and the last five positions average 0.867 to 0.879. An all-token mean therefore weights a short prompt heavily toward its low-entropy opening and a long prompt toward its high-entropy body.

4.3 A position-controlled readout removes the gradient and reverses the extremes

Measured at the final prefill token, routing entropy no longer orders by level (Figure 1, squares): $\rho = +0.02$ ($p = 0.82$) for DeepSeek V3.1 and $\rho = -0.06$ ($p = 0.42$) for Qwen. Its correlation with token count is also small ($\rho = +0.16$ and -0.22), as expected of a readout taken at one position (Table 1).

A direct comparison of the lowest and highest levels shows the same thing as a direction change (Table 2). Under the all-token mean, level 12 sits above level 1 in both models (DeepSeek V3.1, $z = -4.00$, $p = 6.4 \times 10^{-5}$; Qwen, $z = -2.94$, $p = 3.3 \times 10^{-3}$). Under the final-token readout the order flips: level 1 sits above level 12 (DeepSeek V3.1, $z = +3.77$, $p = 1.7 \times 10^{-4}$; Qwen,

Table 2: Level-1 versus level-12 routing entropy (Wilcoxon rank-sum, $n = 14$ per level). The all-token mean places the complex level higher; the final-token readout places the simple level higher.

Model	Readout	L1 mean	L12 mean	Direction (p)
DeepSeek V3.1	all-token	0.837	0.886	L12 > L1 (6.4×10^{-5})
DeepSeek V3.1	final-token	0.870	0.848	L1 > L12 (1.7×10^{-4})
Qwen 397B	all-token	0.881	0.886	L12 > L1 (3.3×10^{-3})
Qwen 397B	final-token	0.884	0.879	L1 > L12 (9.4×10^{-4})

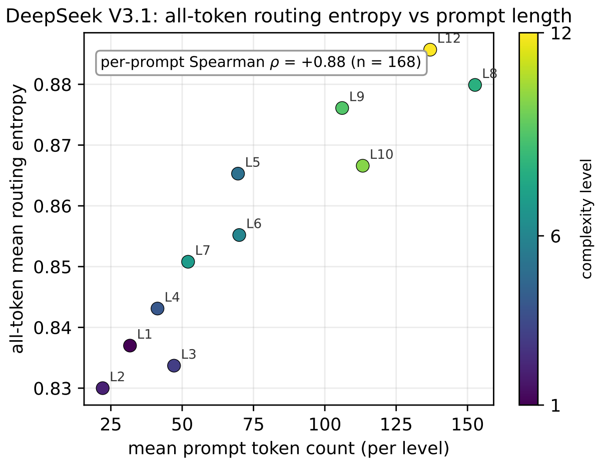

Figure 2: DeepSeek V3.1 per-level all-token mean routing entropy against per-level mean prompt token count; markers labeled and colored by level. The annotated correlation is the per-prompt Spearman value over all 168 prompts.

$z = +3.31$, $p = 9.4 \times 10^{-4}$). At a single position that already covers the whole prompt, the short, low-level prompts carry slightly higher routing entropy than the long, high-level ones; the gap is about 0.022 in DeepSeek V3.1 and 0.006 in Qwen.

4.4 Replication and scope checks

A DeepSeek R1 run on the same 168-prompt suite reproduces the all-token picture: routing entropy correlates with level at $\rho = +0.84$ and with token count at $\rho = +0.86$, again ordering by length at least as well as by level. An earlier 98-prompt baseline on DeepSeek V3.1 showed the same signature on the generation side, where the all-token mean over generated tokens correlated with generated length ($\rho = +0.51$, $p \approx 5 \times 10^{-7}$) more clearly than with level ($\rho = +0.20$, $p = 0.05$). A tightly templated forced-choice suite with a narrow length band (227 to 245 tokens) gives a scoping check in the other direction: on the 17 of 32 prompts recoverable as raw captures, all in the three lowest levels, the all-token mean correlated negatively with both token count ($\rho = -0.66$) and level ($\rho = -0.58$), with token count again accounting for more variance than level.

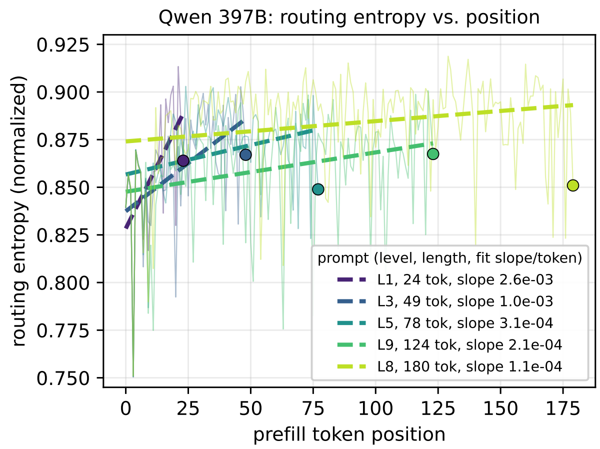

Figure 3: Per-token routing entropy against prefill position for five Qwen3.5-397B prompts (thin lines) with each prompt's least-squares fit (dashed) and final token (dot). Entropy rises with position in every prompt; the mean over positions therefore depends on how many positions there are.

That subset is small and spans three levels, so we read it only as confirmation that the readout remains sensitive to prompt construction when length is nearly fixed.

5 Discussion

Why averaging over positions favors long prompts. Under causal attention the representation at a prefill position depends on every earlier position, so later tokens are computed over more context. In both models the router responds by spreading weight a little more widely at later positions (Section 4, Figure 3). The all-token mean weights every position equally, so it collects more high-entropy positions from a long prompt than from a short one, and when the prompt set is ordered so that the intended-complexity levels also grow longer, the mean rises with level on length alone. The final-token readout avoids the pooling step: it reports one position whose context is the full prompt, so every prompt contributes a single value regardless of length.

The reversal. At the final token the short low-level prompts carry slightly higher routing entropy than the long high-level prompts, consistently across both models. A plain description is that with less prior context to condition on, the router at the final position spreads weight a little more widely. We offer that as the measured direction; the effect is small (about 0.02 and 0.006 in normalized entropy) and its mechanism is not established here.

Implication for routing-entropy comparisons. The practical consequence is a rule for interpretation. A gradient in position-averaged routing entropy across prompts of different lengths is consistent with a pure position effect and therefore cannot, on its own, support a claim that the router or the experts respond to what the prompts demand. This bears directly on readings of router confidence as a difficulty signal (Huang et al., 2024; Raposo et al., 2024): such readings are interpretable between prompts only under a position-controlled readout (the final prefill token, or any fixed position or position-matched window), or under an explicit model of token count and position. The same caution applies to any per-token routing statistic aggregated over inputs of different length, since the positional non-stationarity documented for attention (Xiao et al., 2024) has, on this evidence, a counterpart in the router. Nothing in the mechanism is specific to these models: causal attention gives later positions more context in every decoder-only transformer, and the pattern appeared in every model run here. The finding leaves open whether routing entropy carries information about content, and analyses that hold length fixed or model position explicitly are unaffected by it.

6 Limitations

Intended complexity and prompt length are entangled in this suite, so the design separates position from the all-token average but does not separate length from complexity; whatever level effect might remain at matched length is not estimated here. The final-token readout removes positional averaging at the cost of resting on a single position with its own sampling noise. The position diagnostic covers five prompts on one model. The evidence covers two model families plus an R1 replication, all from quantized weights under one inference setup; other architectures and precisions are untested. The result diagnoses one measurement setup; an all-token average remains useful when position and length are modeled directly.

7 Conclusion

Position-averaged routing entropy rises with intended complexity on this suite because it rises with token position and the complex prompts are longer: the twelve-level gradient follows prompt length at least as closely as it follows the grading, per-token entropy climbs within every prompt examined, and the gradient disappears at the final prefill token, where the extreme levels change places. The same all-token signature appears in DeepSeek V3.1, a DeepSeek R1 replication, and Qwen3.5-397B.

This matters because routing entropy is being read as a signal of model effort, both in interpretations of what MoE experts respond to and in adaptive-computation designs that allocate experts by router confidence (Huang et al., 2024; Raposo et al., 2024). On this suite the aggregation choice determines the sign of the complexity effect, so the same captures support "harder prompts route more diffusely" under one readout and the reverse under another. Any between-prompt claim built on averaged routing entropy, and any comparison of routing statistics across prompt sets whose length distributions differ, therefore inherits a positional artifact large enough to reverse its conclusion. Because the mechanism is not specific to these models, position or length should be treated as a default covariate in routing analyses, and a position-controlled readout, or an explicit model of length and position, is the minimum condition under which a difference in routing entropy can be read as a difference in what the prompts demand.

Data and Code Availability

The measurements in this paper are recomputed from a preserved capture archive (`token-confound-archive/`) by `make_figures.py`, which produces the three figures and the reported correlations from the per-prompt routing-entropy and token-count records and runs no model. The per-level summaries and correlations are transcribed from the archive's evidence summary (`CROSS_MODEL_POSITION_CONFOUND.md`); per-prompt token counts and the DeepSeek R1 values are recomputed from the archived per-prompt JSON, and the position diagnostic from `data/diagnostic_results.json`. A claim-by-claim source index is given in `SOURCES.md`. All reported aggregates are log-derived: an earlier tabulation step in the archive had produced per-prompt rows that matched the correct aggregates without being extracted from the run logs, the tables were regenerated from the logs, and the superseded write-ups are retained in the archive as such. No new model training was performed.

AI Collaboration Statement

Generative AI (Anthropic's Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All captures, statistics, and numerical results are the author's; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content.

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This work received no external funding. The author declares no competing interests. The models studied are third-party releases used as-is for measurement; the author has no affiliation with their creators.

References

Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. arXiv:2401.06066, 2024.

DeepSeek-AI. DeepSeek-V3 technical report. arXiv:2412.19437, 2024.

DeepSeek-AI. DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature*, 645: 633–638, 2025. doi: 10.1038/s41586-025-09422-z.

William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. doi: 10.48550/arXiv.2101.03961.

Quzhe Huang, Zhenwei An, Nan Zhuang, Mingxu Tao, Chen Zhang, Yang Jin, Kun Xu, Liwei Chen, Songfang Huang, and Yansong Feng. Harder tasks need more experts: Dynamic routing in MoE models. In *Findings of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2403.07652.

Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.

Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, Lélio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Théophile Gervet, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mixtral of experts. arXiv:2401.04088, 2024.

David Raposo, Sam Ritter, Blake Richards, Timothy Lillicrap, Peter Conway Humphreys, and Adam Santoro. Mixture-of-Depths: Dynamically allocating compute in transformer-based language models. arXiv:2404.02258, 2024.

Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated Mixture-of-Experts layer. arXiv:1701.06538, 2017.

Jeffrey Shorthill. The generation half: Why prompt-routing studies understate domain specialization in mixture-of-experts models, 2026. Companion manuscript.

Prasann Singhal, Tanya Goyal, Jiacheng Xu, and Greg Durrett. A long way to go: Investigating length correlations in RLHF. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2310.03716.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. arXiv:1706.03762.

Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.17453.

An Yang et al. Qwen3 technical report. arXiv:2505.09388, 2025.